

How Degeneracy Arises from Symmetry in Quantum Mechanics

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1 Introduction

In quantum mechanics, degeneracy refers to the occurrence of multiple distinct quantum states sharing the same energy eigenvalue. A fundamental origin of degeneracy is symmetry. When a system possesses symmetry, it implies the existence of transformations that leave the Hamiltonian invariant. These symmetries lead directly to degeneracies in the energy spectrum.

2 Symmetry in Quantum Mechanics

A symmetry transformation is represented by an operator $\hat{U}$ such that:

$$\hat{U}^\dagger \hat{H} \hat{U} = \hat{H} \quad (1)$$

This implies:

$$[\hat{H}, \hat{U}] = 0 \quad (2)$$

Thus, $\hat{H}$ and $\hat{U}$ share a common set of eigenstates or can be simultaneously diagonalized.

3 Degeneracy from Commuting Operators

Suppose $|\psi\rangle$ is an eigenstate of $\hat{H}$:

$$\hat{H} |\psi\rangle = E |\psi\rangle \quad (3)$$

Apply the symmetry operator $\hat{U}$:

$$\hat{H}(\hat{U}|\psi\rangle) = \hat{U}\hat{H}|\psi\rangle = E(\hat{U}|\psi\rangle) \quad (4)$$

Thus, $\hat{U}|\psi\rangle$ is also an eigenstate with the same energy E .

If $\hat{U}|\psi\rangle$ is linearly independent of $|\psi\rangle$, then degeneracy arises.

4 Continuous Symmetry and Generators

For continuous symmetries, $\hat{U}$ can be written as:

$$\hat{U} = e^{-i\alpha\hat{G}} \quad (5)$$

where $\hat{G}$ is the generator of the symmetry.

If:

$$[\hat{H}, \hat{G}] = 0 \quad (6)$$

then $\hat{G}$ is a conserved quantity.

4.1 Example: Rotational Symmetry

For a spherically symmetric Hamiltonian:

$$[\hat{H}, \hat{L}^2] = 0, \quad [\hat{H}, \hat{L}_z] = 0 \quad (7)$$

Eigenstates are labeled by quantum numbers l and m , and energy depends only on l :

$$E = E_l \quad (8)$$

Since $m = -l, -l+1, \dots, l$, the degeneracy is:

$$g = 2l + 1 \quad (9)$$

5 Group Theoretic Interpretation

Symmetries form groups, and quantum states transform under representations of these groups.

5.1 Irreducible Representations

States belonging to the same irreducible representation (irrep) have the same energy if the Hamiltonian is symmetric under the group.

The dimension of the irrep determines the degeneracy.

6 Example: Hydrogen Atom

The hydrogen atom has a higher symmetry than simple rotational symmetry, associated with the Runge-Lenz vector.

Energy depends only on the principal quantum number n :

$$E_n = -\frac{13.6 \text{ eV}}{n^2} \quad (10)$$

Degeneracy:

$$g = n^2 \quad (11)$$

This enhanced degeneracy arises from hidden symmetry.

7 Symmetry Breaking

If a symmetry is broken (e.g., by an external field), degeneracy is lifted.

7.1 Example: Zeeman Effect

In a magnetic field:

$$\hat{H}' = -\vec{\mu} \cdot \vec{B} \quad (12)$$

The degeneracy in m is removed, splitting energy levels.

8 Conclusion

Degeneracy arises from symmetry because symmetry operations map eigenstates to other states with the same energy. Group theory formalizes this by associating degeneracy with the dimensions of irreducible representations. When symmetry is broken, degeneracy is typically lifted.

9 Multiple Choice Questions

1. Degeneracy means:

- (a) Same state, different energy
- (b) Different states, same energy
- (c) Same operator, different eigenvalues
- (d) No eigenvalues

Answer: b

2. Symmetry implies:

- (a) $[\hat{H}, \hat{U}] \neq 0$
- (b) $[\hat{H}, \hat{U}] = 0$
- (c) $\hat{H} = 0$
- (d) $\hat{U} = 0$

Answer: b

3. Degeneracy arises if:

- (a) $\hat{U} |\psi\rangle = 0$
- (b) $\hat{U} |\psi\rangle$ is linearly independent
- (c) $\hat{U}$ is zero
- (d) $\hat{H}$ is zero

Answer: b

4. Rotational symmetry leads to degeneracy:

- (a) l
- (b) $2l + 1$
- (c) l^2
- (d) n

Answer: b

5. Continuous symmetry generators:

- (a) Do not commute with $\hat{H}$
- (b) Commute with $\hat{H}$
- (c) Are zero
- (d) Are non-Hermitian

Answer: b

6. Hydrogen atom degeneracy is:

- (a) n
- (b) $2n$
- (c) n^2
- (d) l^2

Answer: c

7. Symmetry breaking:

- (a) Increases degeneracy
- (b) Has no effect
- (c) Lifts degeneracy
- (d) Removes eigenstates

Answer: c

8. Degeneracy is related to:

- (a) Temperature
- (b) Group representations
- (c) Classical motion
- (d) Noise

Answer: b